

Question Bank for JAVA

UNIT I

1. Compare abstraction and encapsulation in Java
2. Compare procedure-oriented and object-oriented programming approaches
3. Define polymorphism and state its importance in Java
4. Describe how Java achieves platform independence
5. Develop a basic Java program structure and justify its components
6. Enumerate the essential features of Object-Oriented Programming
7. Evaluate the advantages of object-oriented programming over procedural programming
8. Examine the working of the Java Virtual Machine and bytecode execution process
9. Explain the role of Java in developing Internet-based applications
10. Explain the steps involved in Java environment setup and program execution
11. Identify message communication between objects in Java
12. Justify how encapsulation enhances data security in Java applications
13. Name the concept of polymorphism and its relevance in Java
14. State the characteristics of Object-Oriented Programming
15. State the principle of abstraction in Object-Oriented Programming

UNIT II

1. Analyze decision-making statements available in Java
2. Assess variable scope rules followed in Java programs
3. Classify control statements used in Java with examples
4. Classify Java operators based on their functionality with examples
5. Compare wrapper classes with primitive data types in Java
6. Demonstrate array creation and initialization for primitive data types
7. Demonstrate string concatenation and conversion techniques in Java
8. Differentiate between implicit and explicit type conversion in Java
9. Identify different types of literals supported by Java
10. Investigate the garbage collection process in Java
11. List different types of arrays supported by Java
12. List rules for naming identifiers in Java
13. List various primitive data types supported by Java
14. Show the syntax used for declaring variables and constants in Java
15. Write the syntax of various comment styles used in Java

UNIT III

1. Apply constructor overloading using a suitable Java program
2. Construct a Java program illustrating constructor overloading
3. Construct a Java program using static data members

4. Demonstrate method overloading using a suitable Java program
5. Demonstrate the procedure for creating and accessing objects in Java
6. Distinguish between static members and instance members in Java
7. Evaluate the advantages of constructor overloading
8. Explain the rules for accessing class members using access specifiers in Java with suitable examples.
9. Illustrate how an object can be passed as a parameter to a method in Java using a suitable program.
10. Indicate the purpose of the this keyword in Java
11. Justify the use of method overloading in Java applications with respect to code readability and reusability.
12. Mention the role of constructors in Java programs
13. Recognize the function of the final keyword in Java
14. Specify the role of the static keyword in Java
15. State the significance of classes and objects in Java

UNIT IV

1. Analyze hierarchical inheritance with a suitable example
2. Analyze the limitations of multiple inheritance in Java and explain how interfaces help overcome them.
3. Assess the importance of abstract classes in Java applications
4. Demonstrate method overriding in Java by changing the implementation of a superclass method in a subclass.
5. Differentiate between single and multilevel inheritance in Java
6. Differentiate between this and super keywords in Java
7. Evaluate how method overriding supports runtime polymorphism
8. Evaluate the design choice of using inheritance over object composition in Java programs.
9. Explain how the super keyword is used to access superclass constructors and methods in Java.
10. Explain the mechanism of method overriding in Java
11. Identify different types of inheritance supported in Java
12. List the rules that must be followed while overriding a method in Java.
13. Mention the concept of single and multilevel inheritance
14. Recognize the relationship between superclass and subclass
15. State the purpose of the extends keyword in Java inheritance.

UNIT V

1. Analyze the control flow of a Java program when an exception occurs inside a try block.
2. Assess the necessity of exception handling in Java applications
3. Demonstrate handling of a user-defined exception in Java with a suitable example.
4. Demonstrate handling of multiple exceptions in Java

5. Evaluate the effectiveness of Java's exception handling model in improving program reliability.
6. Explain the role of the finally block in Java exception handling.
7. Identify the difference between checked and unchecked exceptions in Java.
8. Illustrate exception propagation with a suitable Java example
9. Illustrate the use of the throw keyword to explicitly generate an exception in a Java program.
10. Justify the use of custom exceptions over standard exceptions in complex Java applications.
11. List the keywords used in Java for handling exceptions and state their purpose.
12. Mention the meaning of an exception in Java
13. Outline the hierarchy of built-in exception classes in Java
14. State the importance of built-in classes for exception handling
15. Summarize the mechanism of exception handling in Java